
On Variance Reduction in Learning Mean Flows

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Abstract

One-step generative modeling has emerged as a leading approach to amortize the inference cost of diffusion and flow-matching models. Among distillation-free methods, MeanFlow training is notoriously unstable, with non-decreasing loss and unbounded gradient variance. In this work, we establish a theory that attributes this pathology to a misuse of the conditional velocity field: it plays two distinct statistical roles in the loss, both as an unbiased regression target and as a Monte Carlo control variate inside a Jacobi-vector product, with the original loss assigning the wrong coefficient to the latter. We derive the optimal coefficient in closed form, and show that a family of fixes in concurrent works corresponds to different practical realizations of the same optimum. A controlled sweep of this coefficient on two-dimensional benchmarks and on a latent Diffusion Transformer recovers the predicted bias-variance ordering. The optimal coefficient yields up to a 54% improvement in sample quality on two-dimensional benchmarks and a monotone FID trend at every matched-step DiT checkpoint. Crucially, the same DiT measurement also reveals a quantitative *FID-MSE landscape mismatch*: although gradient variance is minimized at an interior coefficient value, the coefficient that minimizes FID prefers the direct use of conditional velocity.

1 Introduction

Deep generative models that leverage neural networks to parametrize and sample from unknown data distributions have achieved remarkable success [1–5]. Among these models, flow-based generative models [6–12] learn a continuous-time transformation between a simple prior and the data distribution. Conditional flow matching [10, 11] and stochastic interpolants [12, 13] further enable simulation-free training by regressing a velocity field, achieving sample quality competitive with diffusion models [1, 14, 2]. However, integration at inference time remains a challenge due to its computational costs and numerical instability, motivating a series of subsequent works on *one-step generative models*, including progressive distillation [15], distribution matching [16, 17], consistency models [18, 19], and flow map matching [20]. Nevertheless, most of them require a pre-trained distillation teacher.

Mean Flows [21] provide a *distillation-free* alternative by learning a two-parameter average velocity field through least-squares regression of a total-derivative identity between the average and instantaneous velocity. In principle, this identity yields exact self-supervision. In practice, however, training is plagued by *non-decreasing loss* and prohibitively *high gradient variance* [22, 23]. A plethora of concurrent works have proposed remedies that target different mechanisms. AlphaFlow [22] attributes slow convergence to gradient conflict between flow-matching and total-derivative consistency components and proposes an α -curriculum. Improved MeanFlow [23] replaces the conditional-velocity tangent with a learned marginal-velocity head. Re-MeanFlow [24] preprocesses the data with rectified-flow straightening to reduce trajectory curvature. Terminal Velocity Matching [25] sidesteps the spatial Jacobian by differentiating with respect to the termi-

nal time. Functional Mean Flows [26] extend MeanFlow to Hilbert spaces and study marginal-conditional consistency for functional data. Each fix is empirically effective in isolation, but no theory explains *why* the original objective is unstable or *what* these fixes have in common.

In this paper, we establish our theory around a single research question:

Why do stochastic tangents destabilize learning Mean Flows?

We answer this question through the lens of *variance reduction*. Specifically, we identify that the conditional velocity plays two statistically distinct roles in the MeanFlow loss. As the regression target, it is an *unbiased* estimator of the marginal velocity. As the tangent inside the Jacobian-vector product (JVP) of the total-derivative identity, however, it acts as a Monte Carlo *control variate* for the inaccessible marginal velocity, with a statistically suboptimal coefficient. The bias-variance trade-off induced by this substitution governs the variance of the gradient through a quadratic term. A Jacobi factor in JVP amplifies sample-based conditional fluctuations at each gradient step, and stop-gradient further hides this amplification from the optimizer, leading to the empirical pathology.

Following the theory, we derive the optimal tangent control-variate coefficient in closed form and show that it converges to a deterministic-tangent estimator in the practical regime where a marginal-velocity estimator is available. This result *explains* why deterministic tangents (*e.g.*, *learned velocity heads*) help stabilize training. They are essentially different practical instantiations of the same statistical optimum. To empirically validate the framework, we perform a controlled sweep of the coefficient across two-dimensional datasets and the ImageNet dataset using latent diffusion Transformers (DiTs). Direct measurement of per-step gradient variance recovers the predicted non-decreasing loss component. Meanwhile, we observe a $1.2\times \sim 4.3\times$ reduction with deterministic tangents on the two-dimensional datasets. The same sweep on ImageNet also reveals a quantitative *FID-MSE landscape mismatch*: the FID ordering aligns with the bias-variance prediction across all four coefficient values. Nonetheless, the FID-axis offset ratio across large coefficients is super-linear in the MSE-axis, hence the FID-optimal coefficient shifts past the gradient-MSE interior minimizer to the *unbiased* corner. Our contributions in this paper are as follows:

- C1** We establish a theory to prove that per-step gradient variance in the original MeanFlow scales unboundedly with the spatial-Jacobi factor in JVP and the stop-gradient operator induces a *semi-gradient gap* that hides this term from the optimizer.
- C2** We frame the JVP tangent as a *control variate* for the marginal velocity, derive the optimal coefficient in closed form, and unify concurrent works by showing that they each correspond to a different practical realization of the optimal coefficient.
- C3** We empirically validate the framework through a controlled sweep of the coefficient on both two-dimensional datasets and ImageNet with latent DiTs. We observe $1.2\times \sim 4.3\times$ direct gradient-variance reduction, up to 54% SW_1 improvements on two-dimensional benchmarks, the predicted bias-variance FID ordering at every matched-step DiT checkpoint, and a quantitative FID-MSE landscape mismatch: the optimal coefficient minimizing gradient-MSE is near 0.94. In contrast, the one that minimizes FID leans toward 0.

2 Preliminaries

Flow-Matching. Let $\mathbf{x}$ denote a data sample in $\mathcal{X} \subset \mathbb{R}^d$. Flow-matching generative models [10, 11] learn a time-dependent diffeomorphism $\psi(\mathbf{x}, t) : \mathbb{R}^d \times [0, 1] \rightarrow \mathbb{R}^d$ from a simple distribution $\mathbf{x}_1 \sim p(\mathbf{x}, 1)$ to the target $\mathbf{x}_0 \sim p(\mathbf{x}, 0) = p_{\text{data}}$ by parameterizing a velocity field $\mathbf{v}(\mathbf{x}, t) \triangleq \frac{d}{dt} \psi(\mathbf{x}, t)$. Since the exact *marginal velocity field* $\mathbf{v}(\mathbf{x}, t)$ is generally inaccessible, conditional flow-matching [10] instead learns a conditional field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$ with $\mathbf{x}_t = (1-t)\mathbf{x}_0 + t\mathbf{x}_1$. The following lemma justifies the substitution. See Appendix B.1 for its proof.

Lemma 1. *Given vector fields $\mathbf{v}_{\text{cond}}$ generating conditional probability paths $p(\mathbf{x}, t | \mathbf{x}_0)$, for any $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, the expected conditional velocity field is equal to the marginal velocity field*

$$\mathbf{v}(\mathbf{x}, t) = \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0) | \mathbf{x}_t = \mathbf{x}}[\mathbf{v}(\mathbf{x}, t | \mathbf{x}_0)]. \quad (1)$$

One-step Mean Flows. Iterative integration of $\mathbf{v}(\mathbf{x}, t)$ at inference is expensive and can be numerically unstable. To bypass it, Mean Flows [21] learn a two-parameter average velocity field $\mathbf{u}(\mathbf{x}, r, t)$

87 through regressing an identity along the flow trajectory $\mathbf{x}_\tau = \psi(\mathbf{x}_r, \tau)$ with terminal at $\mathbf{x}_t = \mathbf{x}$:

$$(t - r) \mathbf{u}(\mathbf{x}, r, t) = \int_r^t \mathbf{v}(\mathbf{x}_\tau, \tau) d\tau. \quad (2)$$

88 Differentiating with respect to the terminal time t along the same trajectory yields

$$\mathbf{u}(\mathbf{x}, r, t) + (t - r) \frac{d}{dt} \mathbf{u}(\mathbf{x}, r, t) = \mathbf{v}(\mathbf{x}, t), \quad \frac{d}{dt} \mathbf{u} = \partial_{\mathbf{x}} \mathbf{u} \cdot \mathbf{v}(\mathbf{x}, t) + \partial_t \mathbf{u}, \quad (3)$$

89 where the total derivative $\frac{d}{dt} \mathbf{u}$ is essentially a JVP, denoted by $\text{JVP}(\mathbf{u}, (\mathbf{x}, r, t), (\mathbf{v}(\mathbf{x}, t), 0, 1))$. The
90 original MeanFlow paper [21] replaces *both* occurrences of the marginal field with the conditional
91 field $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$, giving the MeanFlow loss

$$\mathcal{L}_{\text{MF}}(\boldsymbol{\theta}) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_{\boldsymbol{\theta}}, (\mathbf{x}_t, r, t), (\mathbf{v}_{\text{cond}}, 0, 1))]\|_2^2 \right], \quad (4)$$

92 with $r, t \sim \mathcal{U}(0, 1)$, $r \leq t$, $\mathbf{x}_0 \sim p_{\text{data}}$, $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I}_d)$, $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$, and the stop-gradient
93 $\text{sg}[\cdot]$ avoiding double backpropagation through the JVP.

94 3 Variance Reduction in Learning Mean Flows

95 The MeanFlow loss in equation 4 is empirically known to be *non-decreasing* and to suffer from
96 *high-variance gradients* [22, 23]. We investigate the statistical mechanism behind this pathology.
97 Importantly, our analysis isolates the role of the conditional velocity field $\mathbf{v}_{\text{cond}}$ when it appears
98 as the JVP tangent, and identifies that it acts as a Monte Carlo control variate for the inaccessible
99 marginal velocity, with a coefficient that the original MeanFlow fixes to a suboptimal value. We
100 derive the optimal coefficient and establish a connection to practical fixes in concurrent works.

101 3.1 Two roles of the conditional velocity field

102 With Reynolds decomposition, we express the conditional velocity by $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}, t) + \mathbf{v}'$ with
103 $\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ according to equation 1 and $\Sigma_{\mathbf{v}'} \triangleq \text{Cov}_{\mathbf{x}_0 | \mathbf{x}_t}[\mathbf{v}']$. The MeanFlow loss in equation 4
104 uses $\mathbf{v}_{\text{cond}}$ both as the *regression target* and as the *tangent* inside the total derivative. Carrying the
105 decomposition through per-sample loss $\ell_{\text{MF}}(\boldsymbol{\theta})$ yields:

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t}[\ell_{\text{MF}}(\boldsymbol{\theta})] = \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|_2^2 + \text{sg}[\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)], \quad (5)$$

106 where $\mathbf{J} \triangleq (t - r) \partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$ is a *Jacobi factor* determined by the Jacobian matrix $\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}}$
107 and $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}} \triangleq \mathbf{u}_{\boldsymbol{\theta}} + (t - r) \text{sg}[\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}] - \mathbf{v}$ is the deterministic mean-field residual. Since the
108 residual $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the exact loss we aim to minimize given by equation 3, the trace term leads to the
109 non-decreasing loss and eliminates *iff* the Jacobi satisfies $\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} = \frac{1}{t - r} \mathbf{I}_d$ (equivalently, $\mathbf{J} = \mathbf{0}$). We
110 identify two roles of the conditional velocity field in this decomposition:

111 **Unbiased Target.** As the regression target, $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *linearly* into the residual, contributing
112 the *irreducible* per-sample noise floor $\text{Tr}(\Sigma_{\mathbf{v}'})$. However, since this noise is independent of $\boldsymbol{\theta}$ and
113 $\mathbf{v}'$ is zero-mean, $\mathbf{v}_{\text{cond}}$ can serve as an unbiased target for learning Mean Flows.

114 **Variance Amplifier.** As the JVP tangent, the same $\mathbf{v}_{\text{cond}}$ injects $\mathbf{v}'$ *multiplicatively* through the Ja-
115 cobi factor $\mathbf{J}$, contributing the *amplified* term $\text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top)$. It scales unboundedly with the spectral
116 magnitude of $\mathbf{J}$. This amplification is the dominant driver of variance at the gradient level:

117 **Theorem 2** (Jacobian Variance Amplification). *Let $\mathbf{g} \triangleq \nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_t, r, t) \in \mathbb{R}^{d \times p}$ be the parameter*
118 *Jacobian of the average velocity. The trace of the conditional gradient covariance (i.e., the total*
119 *variance) of $\ell_{\text{MF}}(\boldsymbol{\theta})$ in equation 4 satisfies*

$$\text{Tr}(\text{Cov}[\nabla_{\boldsymbol{\theta}} \ell_{\text{MF}} | \mathbf{x}_t]) \propto \text{Tr}(\mathbf{g}^\top \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J} \mathbf{g}).$$

120

121 See the proof in Appendix B. The theorem identifies that the total gradient variance grows with $\|\mathbf{J}\|^2$
122 for fixed $\mathbf{g}$ and $\Sigma_{\mathbf{v}'}$, saturates only at the irreducible noise floor when $\mathbf{J} \rightarrow \mathbf{0}$, and is *uncontrolled* by
123 any term in the loss due to the use of the stop-gradient in the original MeanFlow. Meanwhile, without
124 the stop-gradient operator, the optimizer could in principle reduce variance by driving $\mathbf{J} \rightarrow \mathbf{0}$. This
125 establishes a *semi-gradient gap*, formally,

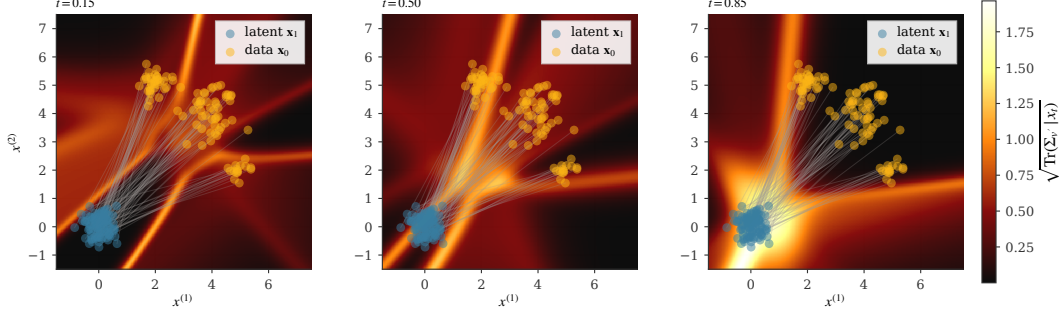


Figure 1: Spatial distribution of $\sqrt{\text{Tr}(\Sigma_{v'} | \mathbf{x}_t)} = \sqrt{\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} \|\mathbf{v}'\|^2}$ at three timesteps on a two-dimensional Gaussian mixture. Conditional variances concentrate in mode-mixing regions.

Theorem 3 (Semi-Gradient Gap). *The gradient of the MeanFlow loss $\mathcal{L}_{MF}$ with and without the stop-gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E} \left[\left(\nabla_{\theta} (\partial_{\mathbf{x}_t} \mathbf{u}_{\theta} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\theta}) \right)^{\top} \mathbf{r}_{\theta} \right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E} [\nabla_{\theta} \text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}. \quad (6)$$

128

In the original MeanFlow, the stop-gradient operator prevents $\mathbf{J}$ from being passed to the optimizer, leading to an empirical non-decreasing loss. Meanwhile, the mean-field difference vanishes at convergence ($\mathbf{r}_{\theta} \rightarrow 0$) and the variance-driven term $\text{Tr}(\mathbf{J} \Sigma_{v'} \mathbf{J}^{\top})$ dominates. To better illustrate the idea, Figure 1 visualizes the spatial distribution of $\sqrt{\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} \|\mathbf{v}'\|^2}$ on a three-mode, two-dimensional Gaussian mixture. It showcases that the total variance magnitude is non-zero, concentrates in mode-mixing regions where conditional paths overlap, and grows toward the latent endpoint ($t \rightarrow 1$).

3.2 Rethinking tangent as a control variate

The amplification in Theorem 2 arises due to using $\mathbf{v}_{\text{cond}} = \mathbf{v} + \mathbf{v}'$ as the JVP tangent injects the zero-mean fluctuation $\mathbf{v}'$. Hence, $\mathbf{v}_{\text{cond}}$ is analogous to the canonical structure of a Monte Carlo control variate [27], where $\mathbf{v}_{\text{cond}}$ is a stochastic estimator of the inaccessible $\mathbf{v}$ with known mean, and the coefficient on the noise term $\mathbf{v}'$ determines the bias-variance trade-off. To make this explicit, we introduce a tangent-mixing coefficient $\beta \in [0, 1]$:

$$\tilde{\mathbf{v}}_{\text{tang}}(\beta) \triangleq (1 - \beta) \mathbf{v}_{\text{cond}} + \beta \hat{\mathbf{v}} = \mathbf{v} + (1 - \beta) \mathbf{v}' + \beta \mathbf{b}, \quad (7)$$

where $\hat{\mathbf{v}}$ can be any deterministic proxy for the marginal velocity (e.g., a velocity network) with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}$ that is deterministic given $\mathbf{x}_t$. Herein, $\beta = 0$ recovers vanilla MeanFlow and $\beta = 1$ replaces the tangent entirely with the deterministic proxy.

Bias-variance trade-off. Substituting $\tilde{\mathbf{v}}_{\text{tang}}(\beta)$ for $\mathbf{v}_{\text{cond}}$ in the JVP tangent and propagating through the loss, the per-sample gradient writes

$$\mathbf{g}^{(\beta)} = 2\mathbf{g}^{\top} \left[\mathbf{r}_{\theta} + ((1 - \beta) \mathbf{J} - \beta \mathbf{I}) \mathbf{v}' + \beta (\mathbf{J} + \mathbf{I}) \mathbf{b} \right], \quad (8)$$

where the noise term scales the conditional fluctuation $\mathbf{v}'$ by the matrix $((1 - \beta) \mathbf{J} - \beta \mathbf{I})$, and the bias term scales the proxy error $\mathbf{b}$ by $\beta (\mathbf{J} + \mathbf{I})$. In this case, we can balance the bias-variance trade-off through solving for the optimal coefficient β^* that minimizes the least-squares between the expected per-sample gradient in equation 8 and the target gradients, given by

$$M(\beta) \triangleq \mathbb{E}_{\mathbf{v}' | \mathbf{x}_t} \left[\left\| \mathbf{g}^{(\beta)} - \nabla_{\theta} \mathbf{r}_{\theta} \right\|_2^2 \right] = \mathbb{E}_{\mathbf{v}' | \mathbf{x}_t} \left[\left\| \mathbf{g}^{(\beta)} - 2\mathbf{g}^{\top} \mathbf{r}_{\theta} \right\|_2^2 \right]. \quad (9)$$

We derive the following theorem.

Theorem 4 (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{v'} \approx \sigma^2 \mathbf{I}_d$, $\mathbf{J} \approx \kappa \mathbf{I}_d$, and the parameter-isotropy approximation $\mathbf{g} \mathbf{g}^{\top} \propto \mathbf{I}_d$ (which reduces M to a per-component MSE; cf. Appendix B),*

$$M(\beta) \propto \beta^2 (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta) \kappa - \beta)^2.$$

Table 1: Concurrent MeanFlow remedies as special cases in our control-variate framework. Each method either drives $\beta \rightarrow 1$ with a deterministic proxy, directly reduces the norm of amplification factor $\|\mathbf{J}\|^2$, or replaces the JVP construction altogether.

Method	Mechanism in our framework	Practical realization
AlphaFlow [22]	avoid high- κ regimes	α -curriculum schedule
Improved MF [23]	deterministic tangent ($\beta \rightarrow 1$)	separate velocity head + sg
Modular MF [29]	deterministic tangent ($\beta \rightarrow 1$)	gradient-modulated loss family
Kim <i>et al.</i> [30]	deterministic tangent ($\beta \rightarrow 1$)	staged training schedule
TVM [25]	remove the spatial Jacobian ($\mathbf{J} \rightarrow \mathbf{0}$)	terminal-time differentiation
Re-MeanFlow [24]	reduce $\ \mathbf{J}\ $	rectified-flow preprocessing
Decoupled MF [31]	bypass JVP construction	flow-map conditioning on later t

154 For $\kappa > 0$, $M(\beta)$ admits a unique minimizer

$$\beta^* = \underbrace{\frac{\kappa}{\kappa+1}}_{\text{noise-cancellation}} \cdot \underbrace{\frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}}_{\text{shrinkage}}, \quad (10)$$

155 with optimum value $M(\beta^*) \propto \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$.

156 See the proof in Appendix B. The two components of β^* play distinct roles: $\kappa/(\kappa+1)$ is the coef-
157 ficient that *cancels the Jacobian-amplified noise* through destructive interference between the $\mathbf{J}\mathbf{v}'$
158 and $\mathbf{I}\mathbf{v}'$ contributions in equation 8, while $\sigma^2 d/(\sigma^2 d + \|\mathbf{b}\|^2)$ is the standard James-Stein shrink-
159 age [28] that pulls β^* toward the unbiased corner when the proxy bias $\|\mathbf{b}\|$ is large. Therefore, as
160 the proxy bias shrinks ($\|\mathbf{b}\|^2 \rightarrow 0$), $\beta^* \rightarrow \kappa/(\kappa+1)$, and as the Jacobian magnitude grows ($\kappa \rightarrow \infty$),
161 $\beta^* \rightarrow 1$. Following the theory, we explain the high variance of gradients in the original MeanFlow
162 and establish a connection to practical fixes in concurrent work.

163 **High-variance gradient with suboptimal coefficient.** The original MeanFlow uses $\beta = 0$ which
164 is optimal only when $\kappa = 0$ (i.e., $\partial_{\mathbf{x}_t} \mathbf{u}_\theta = \frac{1}{t-r} \mathbf{I}_d$) or when no deterministic proxy is available
165 ($\|\mathbf{b}\| = \infty$). Both conditions fail in practice: trained backbones produce $\partial_{\mathbf{x}_t} \mathbf{u}_\theta$ matrices that are
166 spectrally far from $\frac{1}{t-r} \mathbf{I}_d$, and there are multiple viable deterministic proxies for the marginal $\mathbf{v}$.

167 **Concurrent works as control variates.** Our theory unifies several remedies for MeanFlow’s train-
168 ing pathology across concurrent works into a single one-parameter family. Each empirical fix corre-
169 sponds to a different practical realization of the optimum β^* , distinguished by *which proxy* they use
170 for $\mathbf{v}$ and *which value of β* the construction implicitly selects. We summarize them in table 1.

171 4 Related Work

172 **Variance reduction in learning flow-based models.** Several recent works address variance in
173 learning flow-based models from complementary perspectives. Stable Velocity [32] reveals a two-
174 regime variance structure in flow matching and proposes composite conditioning; TPC [33] re-
175 duces variance through temporal pair coupling; and preconditioned flow matching [34] addresses
176 ill-conditioning via anisotropic preconditioning. Bertrand et al. [35] argue that stochastic-target
177 variance is negligible for standard flow matching, consistent with our analysis, since the problem-
178 atic amplification we identify is specific to the MeanFlow JVP, not the flow-matching loss itself.
179 Rectified flows [36, 37] reduce trajectory curvature by iterative straightening, which by Theorem 2
180 reduces $\|\mathbf{J}\|^2$ and hence the amplified noise term.

181 **Control variates and target networks.** The control-variate framework we apply has classical roots
182 in Monte Carlo estimation [38]; it is also the analytical structure underlying target networks in deep
183 RL [39] and self-distillation in self-supervised learning [40]. We make the connection explicit by
184 interpreting the EMA-derived tangent in MeanFlow as a target-network-style proxy that drives the
185 control-variate coefficient toward its optimum.

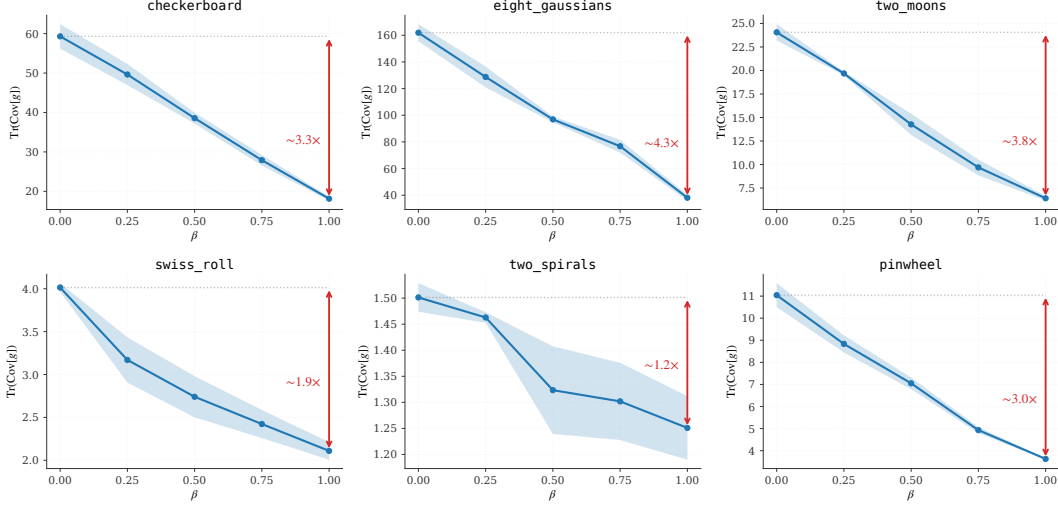


Figure 2: Empirical total gradient variance $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ on six two-dimensional toy datasets with $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$. The monotonic decrease of variances with respect to β on almost every dataset aligns with the prediction in Theorem 2.

5 Experiments

We empirically validate our theory in Section 3 through probing the bias-variance trade-offs. Specifically, we sweep the tangent-mixing coefficient $\beta \in [0, 1]$ end-to-end and compare the results to the closed-form prediction based on Theorem 4.

Instantiation of the $\beta = 1$ corner. To validate the framework empirically, we need a concrete realization of β that we can sweep at scale. We replace the JVP tangent by an exponential moving-average [41] deterministic proxy $\hat{v} \leftarrow \mathbf{u}_{\theta}(\mathbf{x}_t, t, t)$, and use a small flow-matching loss at $r = t$ to keep the proxy bias $\mathbf{b} = \hat{v} - \mathbf{v}$ small and bound the bias term $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ in equation 8. We keep the regression target $\mathbf{v}_{\text{cond}}$ unchanged, exploiting the role asymmetry identified in section 3.1. Appendix C provides details about the full loss, training algorithm, and three propositions characterizing the resulting gradient form, residual bias, and the necessity of the target-tangent role split.

Experiment Setup. We evaluate our theory on three regimes leveraging six two-dimensional toy datasets, dense Gaussian mixtures (DGMM) with various dimensions $d \in \{2, 4, 8, 16, 32, 64\}$, and training DiT-B/4 [42] on ImageNet with Stable Diffusion latents¹. On the 2-D toy and DGMM datasets, we use a three-layer MLP with 128 hidden units, train for 200,000 steps with batch size 256 and three random seeds $\{42, 0, 1\}$, and sweep the eleven-point grid $\beta \in \{0.0, 0.1, \dots, 1.0\}$ for the sample-quality, evaluated by final-step sliced Wasserstein-1 distance (SW_1) on 4096 samples with 500 random projections under a fixed projection seed. On the toy datasets, we additionally run a sub-sweep at $\beta \in \{0, 0.25, 0.5, 0.75, 1\}$ that investigates the per-step total gradient variance $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ using $K = 8$ replica of mini-batches, each with 256 samples, every 2,000 steps and tail-averages over the second half of training. For the DiT, we perform a four-point sweep $\beta \in \{0, 0.25, 0.5, 1\}$ and train the model for 300k steps following the recipe in the original MeanFlow [21]. We defer the full setup, FID table, and direct matrix-form measurement of β^* are to Appendix D.3.

5.1 Variance reduction

Figure 2 reports the total gradient variances $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ on all six 2-D datasets. We observe monotonic decreasing variances with respect to β on almost every dataset. The empirical ordering tracks the variance term $\sigma^2 d((1-\beta)\kappa - \beta)^2$ from Theorem 2 and Theorem 4, empirically validate that the conditional fluctuation $\mathbf{v}'$ is the dominant variance source in the MeanFlow gradients. Meanwhile, we notice that the reduction magnitude varies with the spectral magnitude $\|\mathbf{J}\|$ of the backbone model. The largest reductions appear on the *high-curvature* datasets (e.g., *eight_gaussians*,

¹<https://huggingface.co/enterprise-explorers/sd-vae-ft-mse-flax>

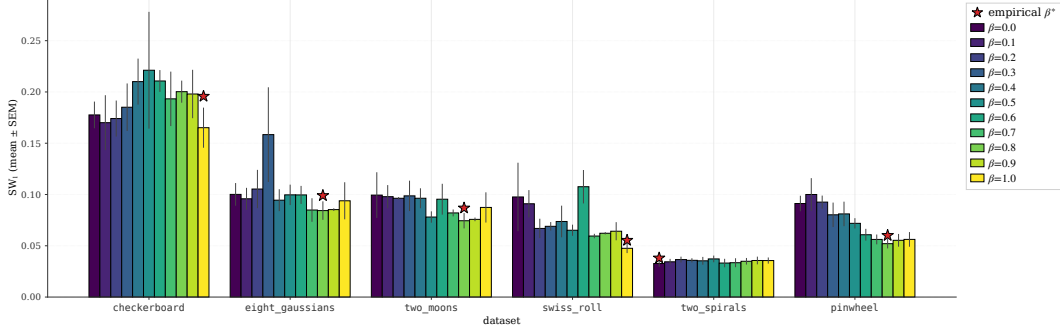


Figure 3: Empirical sample-quality measured by sliced Wasserstein-1 distance $SW_1(\beta)$ on six 2-D toy datasets with $\beta \in \{0.0, 0.1, \dots, 1.0\}$. Bars are grouped by dataset and colored by β with standard error bar. The red star marks the empirical optimal coefficient β^* . Five of six datasets settle at $\beta^* \in \{0.8, 1.0\}$ near the deterministic-tangent corner; `two_spirals` alone settles at $\beta^* = 0$.

Table 2: DGMM β -sweep summary across dimensions $d \in \{2, 4, 8, 16, 32, 64\}$. The quality gap column marks reduction that exceed one standard error mean with $\dagger$. SW_1 is essentially β -insensitive at $d \leq 32$, while only $d = 64$ shows a tentative interior minimum.

d	$SW_1(\beta=0)$	$SW_1(\beta = \beta^*)$	Quality Gap	β^*
2	0.152 ± 0.022	0.146 ± 0.015	$\sim 4\%$	1.0
4	0.083 ± 0.011	0.083 ± 0.011	$< 1\%$	0.8
8	0.062 ± 0.003	0.062 ± 0.003	$\sim 1\%$	1.0
16	0.044 ± 0.002	0.044 ± 0.003	$< 1\%$	1.0
32	0.034 ± 0.001	0.033 ± 0.001	$< 1\%$	0.1
64	0.028 ± 0.003	0.025 ± 0.001	$\sim 9\%^\dagger$	0.4

checkerboard), and the smallest (1.20 $\times$) on `two_spirals`. This is consistent to our theory: low geometric curvature pushes the network toward the flat regime $\partial_{\mathbf{x}_t} \mathbf{u}_\theta \rightarrow \frac{1}{t-r} \mathbf{I}_d$, corresponding to $\kappa \approx 0$, and our theorem 4 predicts both a small per-step variance reduction and a left-shift of optimal coefficient $\beta^* \rightarrow 0$ given by the noise-cancellation factor $\kappa/(\kappa+1)$.

5.2 Bias-variance trade-offs

Whereas $\text{Tr}(\text{Cov}[\nabla_{\theta} \ell_{\text{MF}}])$ reveals the variance reduction induced by β , the sample-quality measured by $SW_1(\beta)$ exposes the full bias-variance trade-off. Figure 3 traces $SW_1(\beta)$ across the full 11-point grid on six 2-D toy datasets. On the five high-curvature datasets, the empirical β^* falls in between $\{0.8, 1.0\}$, recovering the predicted $\beta \rightarrow 1$ location with a small bias norm $\|\mathbf{b}\|_2^2$ when the EMA proxy is accurate. On the lower-curvature `two_spirals` dataset, consistent with section 5.1, the geometry drives $\partial_{\mathbf{x}_t} \mathbf{u}_\theta \rightarrow \frac{1}{t-r} \mathbf{I}_d$, essentially pushing κ toward zero. The predicted β^* then shrinks toward the unbiased corner $\beta = 0$ through the noise-cancellation factor $\kappa/(\kappa+1)$ in Theorem 4, and the empirical sweep result concurs. Meanwhile, we are surprised that the $\beta^* = 1$ instantiation reduces SW_1 by 54% compared to the original MeanFlow ($\beta = 0$) on the `swiss_roll` dataset.

5.3 Scaling with feature dimensions

We investigate whether our theory holds with increasing in feature dimensions by sweeping the coefficient $\beta \in \{0, 0.1, \dots, 1\}$ on dense Gaussian mixture (DGMM) datasets in $\mathbb{R}^d$, varying $d \in \{2, 4, 8, 16, 32, 64\}$. Table 2 summarizes the sample-quality gap between the original MeanFlow with $\beta=0$ and the empirical β^* at each d . We report full results in Table 5 of Appendix D. On low-dimensional datasets with $d \in \{2, 4, 8, 16, 32\}$, SW_1 is essentially β -insensitive, where the quality gap between $\beta=0$ and β^* stays within one standard error of the mean (SEM). This is consistent with the prediction from theorem 4 when the EMA proxy yields small $\|\mathbf{b}\|_2^2$. The noise-cancellation factor $\kappa/(\kappa+1)$ saturates near 1 across the whole β range, leading to flat $SW_1(\beta)$. At $d=64$ the data hints at an interior minimum at $\beta^* = 0.4$ with a $\sim 9\%$ reduction over $\beta=0$, consistent with κ coming off

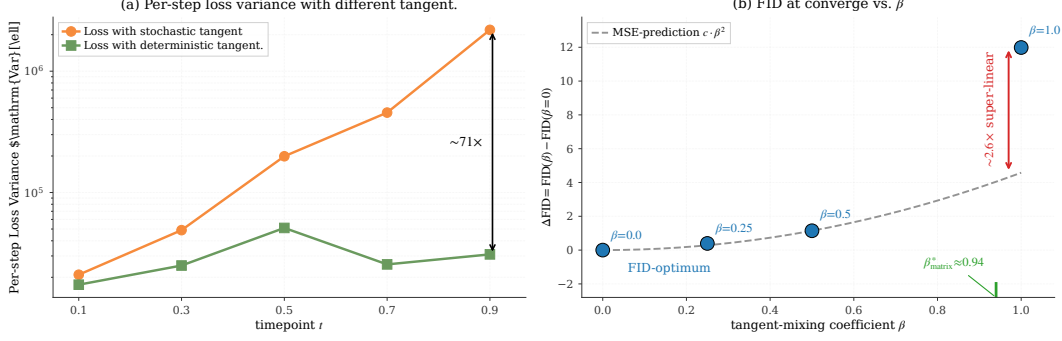


Figure 4: Experiment results training DiT-B/4 on ImageNet-256. *Left*: Per-step loss variance under the stochastic ($\beta = 0$) versus deterministic ($\beta = 1$) tangent, measured on the baseline ($\beta = 0$) checkpoint with the same data and noise. The deterministic-tangent variance is roughly t -flat while the stochastic-tangent variance grows by two orders of magnitude with t , confirming the Jacobi-factor amplification of Theorem 2. *Right*: FID at converge with respect to different β . We report empirical $\Delta\text{FID}(\beta) = \text{FID}(\beta) - \text{FID}(\beta = 0)$ at the matched training step 295k with different β . The dashed gray curve is the quadratic $c\beta^2$. The FID at converge with $\beta = 1$ sits a factor of ~ 2.6 above the quadratic curve, indicating a super-linear relationship between the FID and the bias. The optimal coefficient calculated in matrix form is $\beta_{\text{matrix}}^* \approx 0.94$.

saturation with increasing data dimension d . In the following section 5.4, we will showcase that the same observation is consistent with larger and more complicated DiT model, where $\beta_{\text{matrix}}^* \approx 0.94$ also sits short of the deterministic-tangent corner.

5.4 Latent Diffusion Transformers on ImageNet

We further investigate whether our theory generalizes to large model with high-dimensional features. Figure 4 shows two complementary measurements from the experiments with DiT-B/4 on ImageNet. The left panel in Figure 4 validates that the gradient-variance reduction predicted by Theorem 2 carries over from 2-D toy benchmark datasets to DiT. Herein, we directly measure the per-step loss variance using the model trained with the original MeanFlow loss, using two distinct tangents: the stochastic conditional-velocity tangent $\mathbf{v}_{\text{cond}}$ and the deterministic EMA tangent $\mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)$. We observe that the loss variance with deterministic-tangent stays roughly flat across t , while the stochastic-tangent variance grows by two orders of magnitude. The widening gap directly validates the Jacobi-factor amplification of Theorem 2, where the stochastic tangent passes the conditional-velocity noise through $\mathbf{J} = (t-r)\partial_{\mathbf{x}_t}\mathbf{u} - \mathbf{I}$, which the deterministic tangent eliminates by construction. We observe the same pattern using checkpoint of the $\beta = 1$ instantiation (see appendix D.3), confirming that the amplification is a property of the architecture and data.

The right panel in Figure 4 visualizes the FID at converge as a function of β , with y -axis the empirical $\Delta\text{FID}(\beta) \triangleq \text{FID}(\beta) - \text{FID}(\beta = 0)$. The result shows that an ordering of $\text{FID}(\beta = 0) < \text{FID}(\beta = 0.25) < \text{FID}(\beta = 0.5) < \text{FID}(\beta = 1)$ holds at matched-step checkpoints. We report per-checkpoint values in Table 6 of Appendix D.3. The result establishes the following observation:

FID-MSE landscape mismatch. We report the direct measurement of both ingredients in Theorem 4 on the DiT checkpoints, including the EMA-tracking proxy $\|\widehat{\mathbf{b}}\|^2$ and the irreducible noise floor $\sigma^2 d$, in Appendix D.3. The result predicts an optimal β^* in the interior range $[0.46, 0.50]$ under the scalar-isotropic approximation, while the matrix-form β_{matrix}^* is even larger (see below). The empirical ordering of FID at convergence is hence consistent with the prediction. To test whether FID tracks the gradient-MSE *quantitatively*, we fit a MSE-prediction curve $c\beta^2$ and compare the $\beta = 1$ offset, where c is given by

$$c \leftarrow \arg \min_{c \in \mathbb{R}} \|\Delta\text{FID}(\beta) - c\beta^2\|_2^2, \quad \beta \in \{0.0, 0.25, 0.5\}. \quad (11)$$

Figure 4 shows a $\sim 2.6\times$ super-linear amplification along the ΔFID axis, indicating that the FID over-penalizes bias relative to gradient MSE. As a consequence, the FID-optimal β shifts past the gradient-MSE interior minimizer all the way to the unbiased corner $\beta = 0$.

271 To verify that our isotropic approximation in theorem 4 is not artifactual, we directly measured a
 272 matrix-form upper bound on β_{matrix}^* on the baseline checkpoint by sampling actual $\mathbf{v}'$ values from
 273 the conditional distribution (see Appendix B). The estimator omits the bias term $\|(\mathbf{J}+\mathbf{I})\mathbf{b}\|^2$ from
 274 the denominator, giving

$$\beta_{\text{no bias}}^* \triangleq \text{Tr}(\mathbf{J}\Sigma_{\mathbf{v}'}(\mathbf{J}+\mathbf{I})^\top) / \text{Tr}((\mathbf{J}+\mathbf{I})\Sigma_{\mathbf{v}'}(\mathbf{J}+\mathbf{I})^\top) \geq |\beta_{\text{matrix}}^*|, \quad (12)$$

275 since the omitted bias term is *non-negative*. We evaluate the bound at $t \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$
 276 with fixed gap $t - r = 0.25$, and 512 samples per t using $\mathbf{u}_\theta(\mathbf{x}_t, t, t)$ as the marginal-velocity
 277 proxy. The numerator is *strictly positive* at every probed t , ruling out the unbiased-corner regime
 278 $\beta_{\text{matrix}}^* \leq 0$. Aggregated across t , we have $\beta_{\text{no bias}}^* \approx 0.94$, which is close to the deterministic-tangent
 279 corner and *larger* than the scalar-isotropic estimate $\beta^* \in [0.46, 0.50]$. The full β_{matrix}^* then lies
 280 in $(0, 0.94]$. Herein, we do not pin down the exact value due to the inaccessible true model bias
 281 $\mathbf{b} = \mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)$ on ImageNet without access to the conditional distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$.
 282 The matrix-form measurement further strengthens, rather than weakens, the FID-MSE mismatch:
 283 the gradient-MSE optimum sits in $(0, 0.94]$ while the FID optimum sits at the unbiased corner $\beta = 0$.

284 6 Conclusion

285 In this paper, we establish a theory from the perspective of Monte Carlo control variates that isolates
 286 the statistical structure of training pathology in the original MeanFlow. We identify two distinct
 287 roles the conditional velocity plays in the loss, and the original objective assigns the wrong coef-
 288 ficient to one of them. Following the theory, we derive the optimal tangent-mixing coefficient for
 289 the control variate in closed form. In our experiment, we conduct a controlled sweep of the coeffi-
 290 cient to recover the predicted bias-variance trade-offs, scaling from two-dimensional toy datasets to
 291 a large-scale DiT-B/4 model on ImageNet-256 scale. In addition, our experiment with DiT reveals
 292 a quantitative *FID-MSE landscape mismatch*, where the optimal coefficient given by minimizing
 293 matrix-form MSE sits at $\beta^* \approx 0.94$ while the optimal coefficient minimizing the FID leans towards
 294 the unbiased corner $\beta = 0$, with a super-linear relationship between the empirical FID($\beta = 1$) and
 295 the bias. We discuss practical insights and limitations below.

296 **Practical insights.** Our Theorem 4 predicts two factors that determines the empirical β^* through
 297 the data-dependent κ and model bias $\|\mathbf{b}\|^2$. Selecting the appropriate β reflects the balance between
 298 bias and variance. Thereby, when gradient stability is the primary concern, we recommend toggling
 299 β toward the corner $\kappa/(\kappa+1)$, which is attainable through an EMA proxy $\mathbf{u}_\theta(\mathbf{x}, t, t)$ with $O(1)$.
 300 When sample quality is the primary metric, our DiT experiment suggests toggling β back toward 0
 301 or annealing β with respect to $\|\mathbf{b}\|_2^2$ until the FID bias penalty becomes negligible. A metric-aware β
 302 schedule that interpolates between these regimes during training is a promising direction for future
 303 work.

304 **Limitations.** We acknowledge several limitations of this work: (i) *Scalar-isotropic approximation*.
 305 Our Theorem 4 is derived under $\mathbf{J} \approx \kappa \mathbf{I}_d$ and $\Sigma_{\mathbf{v}'} \approx \sigma^2 \mathbf{I}_d$. In the full diagonal matrix-valued set-
 306 ting the optimum is direction-dependent: a per-direction $\beta_i = \frac{\mathbf{J}_{ii}}{\mathbf{J}_{ii}+1} \cdot \frac{\sigma_i^2}{\sigma_i^2+b_i^2}$ would minimize the
 307 per-component gradient MSE, and a single global β trades off residual variance across directions.
 308 This assumption may be valid for backbones whose Jacobian spectrum is concentrated, but it may
 309 fail for more complex models or highly anisotropic data. (ii) *FID-aware loss design*. Our exper-
 310 iment characterizes the FID-MSE mismatch but does not derive an FID-aware loss or β schedule,
 311 which remains an open question. (ii) *Non-Euclidean data*. We conduct our experiment in the Eu-
 312 clidean space $\mathbb{R}^d$. Extending the closed-form β^* to manifold MeanFlow [43] is straightforward at
 313 the gradient level but requires care in defining the second moment $\text{Tr}(\Sigma_{\mathbf{v}'})$ on tangent spaces.

314 **Future work.** There are three promising future directions. First, we argue that our control-variate
 315 diagnosis applies to any self-supervised one-step generative model that bootstraps a parametrized
 316 map as the JVP-style total derivative in MeanFlow, including consistency models [18] and shortcut
 317 models [44]. Meanwhile, a per-sample scalar coefficient driven by online estimates of κ and $\|\mathbf{b}\|_2^2$
 318 can potentially close the global-vs-per-sample and the scalar-vs-matrix gap during training. Last but
 319 not least, an adaptive schedule that interpolates between the gradient-MSE optimum and the FID-
 320 optimal corner, calibrated by either a learned proxy or proxy-quality validation FID, could resolve
 321 the FID-MSE mismatch we identified.

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762 not be described.

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Table 3 lists all the notations we use in this work.

Table 3: Table of notations.

Notation	Description
<i>States and velocities</i>	
$\mathbf{x}, \mathbf{x}_t$	Random data sample / state at interpolant time t
$\mathbf{x}, \mathbf{x}_t$	Realized state / state at time t : $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
$\mathbf{v}(\mathbf{x}, t)$	Marginal velocity field
$\mathbf{v}_{\text{cond}}$	Conditional velocity: $\mathbf{v}_{\text{cond}} = \mathbf{x}_1 - \mathbf{x}_0$
$\mathbf{v}'$	Conditional velocity fluctuation: $\mathbf{v}' = \mathbf{v}_{\text{cond}} - \mathbf{v}(\mathbf{x}_t, t)$, $\mathbb{E}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$
$\hat{\mathbf{v}}$	Deterministic proxy for $\mathbf{v}$ (e.g., EMA velocity at boundary $r = t$)
<i>Networks</i>	
$\mathbf{u}_{\theta}(\mathbf{x}, r, t)$	Learned two-parameter average velocity field (parameters θ)
$\mathbf{u}_{\hat{\theta}}$	EMA copy of $\mathbf{u}_{\theta}$, parameters $\hat{\theta}$
<i>Spatial-Jacobian quantities</i>	
$\partial_{\mathbf{x}_t} \mathbf{u}_{\theta}$	Spatial Jacobian of $\mathbf{u}_{\theta}$ w.r.t. its first argument
$\mathbf{J}$	Jacobi factor: $\mathbf{J} = (t - r)\partial_{\mathbf{x}_t} \mathbf{u}_{\theta} - \mathbf{I}_d \in \mathbb{R}^{d \times d}$
$\mathbf{A}$	Shorthand $\mathbf{A} \triangleq \mathbf{J} + \mathbf{I}_d = (t - r)\partial_{\mathbf{x}_t} \mathbf{u}_{\theta}$
κ	Scalar-isotropic surrogate for $\mathbf{J}$: $\mathbf{J} \approx \kappa \mathbf{I}_d$
<i>Bias-variance ingredients</i>	
$\Sigma_{\mathbf{v}'}$	Conditional velocity covariance: $\text{Cov}_{\mathbf{x}_0 \mathbf{x}_t}[\mathbf{v}']$
$\sigma^2 d$	Trace of $\Sigma_{\mathbf{v}'}$; in scalar-isotropic case $\Sigma_{\mathbf{v}'} = \sigma^2 \mathbf{I}_d$
$\mathbf{b}$	Proxy bias: $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$
$\ \widehat{\mathbf{b}}\ ^2$	EMA-tracking proxy: $\mathbb{E}_{\mathbf{x}_t} [\ \mathbf{u}_{\theta}(\mathbf{x}_t, t, t) - \mathbf{u}_{\hat{\theta}}(\mathbf{x}_t, t, t)\ ^2]$
<i>Gradients</i>	
$\nabla_{\theta} \ell_{\text{MF}}$	Per-step gradient of the MeanFlow loss
$\mathbf{g}$	Parameter Jacobian of the network output: $\mathbf{g} \triangleq \nabla_{\theta} \mathbf{u}_{\theta} \in \mathbb{R}^{d \times p}$
$\mathbf{G}_{\theta}$	Parameter-space Gram matrix: $\mathbf{G}_{\theta} \triangleq \mathbf{g}\mathbf{g}^{\top} \in \mathbb{R}^{d \times d}$
<i>Tangent-mixing coefficient</i>	
$\beta \in [0, 1]$	Tangent-mixing coefficient equation 7
β^*	Closed-form scalar-isotropic minimizer of $M(\beta)$ in Theorem 4
β^*_{matrix}	Matrix-form minimizer (Appendix B, equation equation 22)
$M(\beta)$	Conditional MSE of the per-sample gradient equation 8 at coefficient β
<i>Loss components</i>	
$\mathcal{L}_{\text{MF}}$	Vanilla MeanFlow loss equation 4
$\mathcal{L}_{\text{MF}}^{\text{EMA}}$	EMA-tangent variant of the MeanFlow loss equation 28
$\mathcal{L}_{\text{FM}}$	Flow-matching anchor loss equation 29
$\mathcal{L}_{\beta=1}$	$\beta = 1$ corner training loss equation 30
<i>Operators</i>	
$\text{sg}[\cdot]$	Stop-gradient operator
$\text{JVP}(\mathbf{u}, \mathbf{x}, \mathbf{v})$	Jacobian-vector product of $\mathbf{u}$ at $\mathbf{x}$ in tangent direction $\mathbf{v}$
$\text{Tr}(\cdot), \text{Cov}[\cdot], \text{Var}[\cdot]$	Trace, covariance, variance

780 B Proofs

781 B.1 Proof of Marginal-Velocity Lemma

782 **Lemma 1.** *Given conditional vector fields $\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)$ that generate probability paths $p(\mathbf{x}, t \mid \mathbf{x}_0)$,
783 for any data distribution $\mathbf{x}_0 \sim p(\mathbf{x}_0)$, define the marginal velocity field as*

$$\mathbf{v}(\mathbf{x}, t) \triangleq \mathbb{E}_{\mathbf{x}_0 \sim p(\mathbf{x}_0 \mid \mathbf{x}_t = \mathbf{x})}[\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)].$$

784 Then $\mathbf{v}(\mathbf{x}, t)$ generates the marginal probability path $p(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0$ via the
785 continuity equation.

786 *Proof.* We follow the standard flow-matching construction [1, 2]: the marginal velocity field is
787 defined by the displayed conditional expectation, and it remains to verify that this definition makes
788 $\mathbf{v}(\mathbf{x}, t)$ generate $p(\mathbf{x}, t)$ via the continuity equation. By the law of total probability and the definition
789 of the conditional velocity field,

$$p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t) = \int p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) d\mathbf{x}_0. \quad (13)$$

790 Each conditional pair $(\mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0), p(\mathbf{x}, t \mid \mathbf{x}_0))$ satisfies the continuity equation $\partial_t p(\mathbf{x}, t \mid \mathbf{x}_0) +$
791 $\text{div}(p(\mathbf{x}, t \mid \mathbf{x}_0) \mathbf{v}(\mathbf{x}, t \mid \mathbf{x}_0)) = 0$ by assumption [3]. Integrating against $p(\mathbf{x}_0)$ and exchanging
792 integration with ∂_t and div yields $\partial_t p(\mathbf{x}, t) + \text{div}(p(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)) = 0$, so the constructed $\mathbf{v}(\mathbf{x}, t)$
793 generates $p(\mathbf{x}, t)$. Dividing the displayed equation by $p(\mathbf{x}, t)$ and applying Bayes' rule $p(\mathbf{x}_0 \mid$
794 $\mathbf{x}_t = \mathbf{x}) = p(\mathbf{x}, t \mid \mathbf{x}_0) p(\mathbf{x}_0) / p(\mathbf{x}, t)$ recovers the conditional-expectation form. We treat this as
795 the operating definition of the marginal velocity used throughout; uniqueness up to divergence-free
796 vector fields, which we do not require, is a separate question we do not address. $\square$

797 B.2 Proof of Theorem 2 (Jacobian Variance Amplification)

798 **Theorem 2** (Jacobian Variance Amplification). *The Jacobi factor amplifies the gradient variance
799 of the original MeanFlow loss:*

$$\text{Var}[\nabla_{\boldsymbol{\theta}} \ell_{MF}(\boldsymbol{\theta}) \mid \mathbf{x}_t] \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top).$$

800 *Proof.* Under the stop-gradient operator, the compound prediction $V_{\boldsymbol{\theta}}^{\text{sg}}$ depends on $\boldsymbol{\theta}$ only through
801 $\mathbf{u}_{\boldsymbol{\theta}}(\mathbf{x}_t, r, t)$. Let $\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}} \in \mathbb{R}^{d \times p}$ denote the parameter Jacobian of the network output. The per-step
802 gradient is

$$\nabla_{\boldsymbol{\theta}} \ell_{MF} = 2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top (\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}} + \mathbf{J} \mathbf{v}'), \quad (14)$$

803 where $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ is the mean-field residual and $\mathbf{J} = (t - r) \partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} - \mathbf{I}_d$ is the Jacobi factor. All three
804 quantities $(\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}}, \mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}, \text{ and } \mathbf{J})$ are deterministic conditioned on $\mathbf{x}_t$.

805 Since $\mathbb{E}_{\mathbf{x}_0 \mid \mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$ by equation 1, the conditional mean gradient is:

$$\mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \mathbf{x}_t] = 2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top \mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}. \quad (15)$$

806 The centered gradient is $\nabla_{\boldsymbol{\theta}} \ell - \mathbb{E}[\nabla_{\boldsymbol{\theta}} \ell \mid \mathbf{x}_t] = 2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top \mathbf{J} \mathbf{v}'$, so the conditional variance is:

$$\begin{aligned} \text{Var}[\nabla_{\boldsymbol{\theta}} \ell \mid \mathbf{x}_t] &= \mathbb{E}[\|2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top \mathbf{J} \mathbf{v}'\|^2 \mid \mathbf{x}_t] \\ &= 4 \mathbb{E} \left[\left(\mathbf{v}' \right)^\top \mathbf{J}^\top \underbrace{(\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top}_{\mathbf{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}} \mathbf{J} \mathbf{v}' \mid \mathbf{x}_t \right] \\ &= 4 \text{Tr}(\mathbf{G}_{\boldsymbol{\theta}} \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top) \propto \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \end{aligned} \quad (16)$$

807 where $\mathbf{G}_{\boldsymbol{\theta}} = (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})(\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^\top \in \mathbb{R}^{d \times d}$ is the parameter-space Gram matrix and the last step uses
808 $\mathbf{G}_{\boldsymbol{\theta}} \succeq \mathbf{0}$ to establish proportionality via the cyclic property of the trace. $\square$

809 B.3 Proof of Theorem 3 (Semi-Gradient Gap)

810 **Theorem 3** (Semi-Gradient Gap). *The gradients of the MeanFlow loss with and without the stop-*
811 *gradient operator differ by*

$$\underbrace{2(t-r) \mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} \left[(\nabla_{\boldsymbol{\theta}} (\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_{\boldsymbol{\theta}}))^{\top} \mathbf{r}_{\boldsymbol{\theta}} \right]}_{\text{mean-field gradient difference}} + \underbrace{\mathbb{E}_{r,t,\mathbf{x}_0,\mathbf{x}_1} [\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})]}_{\text{variance-driven gradient difference}}.$$

812 *Proof.* The MeanFlow loss conditioned on $\mathbf{x}_t$ decomposes as (cf. equation 5):

$$\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} [\ell_{\text{MF}}] = \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|^2 + \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (17)$$

813 **Without stop-gradient.** Both terms depend on $\boldsymbol{\theta}$ (through $\mathbf{u}_{\boldsymbol{\theta}}$ and $\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}}$), so the full gradient is:

$$\nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] = \nabla_{\boldsymbol{\theta}} \|\mathbf{r}_{\boldsymbol{\theta}}\|^2 + \nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}). \quad (18)$$

814 **With stop-gradient.** The JVP terms in $\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}$ and $\mathbf{J}$ are treated as constants, so:

$$\nabla_{\boldsymbol{\theta}}^{\text{sg}} \|\mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}\|^2 = 2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^{\top} \mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}, \quad \nabla_{\boldsymbol{\theta}}^{\text{sg}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top}) = \mathbf{0}. \quad (19)$$

815 The gap between the full and stop-gradient gradients is therefore:

$$\begin{aligned} & \nabla_{\boldsymbol{\theta}} \mathbb{E}[\ell] - \nabla_{\boldsymbol{\theta}}^{\text{sg}} \mathbb{E}[\ell] \\ &= \underbrace{\nabla_{\boldsymbol{\theta}} \|\mathbf{r}_{\boldsymbol{\theta}}\|^2 - 2 (\nabla_{\boldsymbol{\theta}} \mathbf{u}_{\boldsymbol{\theta}})^{\top} \mathbf{r}_{\boldsymbol{\theta}}^{\text{sg}}}_{\text{mean-field gradient difference}} + \underbrace{\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})}_{\text{variance-driven difference}}. \end{aligned} \quad (20)$$

816 Expanding the first term: $\nabla_{\boldsymbol{\theta}} \|\mathbf{r}_{\boldsymbol{\theta}}\|^2 = 2(\nabla_{\boldsymbol{\theta}} \mathbf{r}_{\boldsymbol{\theta}})^{\top} \mathbf{r}_{\boldsymbol{\theta}}$, and $\nabla_{\boldsymbol{\theta}} \mathbf{r}_{\boldsymbol{\theta}}$ includes the derivative of the JVP
817 terms $(t-r)(\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}})$ w.r.t. $\boldsymbol{\theta}$, yielding the second-order mean-field difference $2(t-r)$
818 $r) \mathbb{E}[(\nabla_{\boldsymbol{\theta}} (\partial_{\mathbf{x}_t} \mathbf{u}_{\boldsymbol{\theta}} \cdot \mathbf{v} + \partial_t \mathbf{u}_{\boldsymbol{\theta}}))^{\top} \mathbf{r}_{\boldsymbol{\theta}}]$. At convergence $\mathbf{r}_{\boldsymbol{\theta}} \rightarrow \mathbf{0}$, this term vanishes, and the gap is
819 dominated by the variance-driven term $\nabla_{\boldsymbol{\theta}} \text{Tr}(\mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^{\top})$. $\square$

820 B.4 Proof of Theorem 4 (Optimal Control-Variate Coefficient)

821 **Theorem 4** (Optimal control-variate coefficient). *Under the scalar-isotropic approximation $\Sigma_{\mathbf{v}'} \approx$*
822 *$\sigma^2 \mathbf{I}_d$ and $\mathbf{J} \approx \kappa \mathbf{I}_d$, the conditional MSE of $\mathbf{g}^{(\beta)}$ in equation 8 is*

$$M(\beta) = \beta^2 (\kappa+1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1-\beta)\kappa - \beta)^2,$$

823 *with unique minimizer $\beta^* = \frac{\kappa}{\kappa+1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}$ and optimum value $M(\beta^*) = \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$.*
824 *For all $\kappa, \|\mathbf{b}\|^2 > 0$, the optimum strictly dominates both $M(0) = \sigma^2 \kappa^2 d$ and $M(1) = (\kappa+1)^2 \|\mathbf{b}\|^2 +$*
825 *$\sigma^2 d$.*

826 *Proof. Step 1: bias-variance decomposition.* From equation 8,

$$\mathbf{g}^{(\beta)} = 2\mathbf{g}^{\top} \left[\bar{\mathbf{r}} + ((1-\beta)\mathbf{J} - \beta\mathbf{I})\mathbf{v}' + \beta(\mathbf{J} + \mathbf{I})\mathbf{b} \right].$$

827 Using $\mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t}[\mathbf{v}'] = \mathbf{0}$, the conditional mean is $\mathbb{E}[\mathbf{g}^{(\beta)} | \mathbf{x}_t] = 2\mathbf{g}^{\top} (\bar{\mathbf{r}} + \beta(\mathbf{J} + \mathbf{I})\mathbf{b})$ and the centered
828 noise is $-2\mathbf{g}^{\top} ((1-\beta)\mathbf{J} - \beta\mathbf{I})\mathbf{v}'$ (relative to the ideal target $2\mathbf{g}^{\top} \bar{\mathbf{r}}$). The conditional MSE w.r.t. the
829 ideal target therefore decomposes as

$$\text{MSE} = \underbrace{4\beta^2 \|\mathbf{g}^{\top} (\mathbf{J} + \mathbf{I})\mathbf{b}\|^2}_{\text{bias}^2} + \underbrace{4 \text{Tr}(\mathbf{g} \mathbf{g}^{\top} ((1-\beta)\mathbf{J} - \beta\mathbf{I}) \Sigma_{\mathbf{v}'} ((1-\beta)\mathbf{J} - \beta\mathbf{I})^{\top})}_{\text{variance}}.$$

830 **Step 2: matrix-form expansion and strict convexity.** Let $\mathbf{A} \triangleq \mathbf{J} + \mathbf{I}$ and $\mathbf{G}_{\boldsymbol{\theta}} \triangleq \mathbf{g} \mathbf{g}^{\top} \succeq \mathbf{0}$. Substi-
831 tuting $((1-\beta)\mathbf{J} - \beta\mathbf{I}) = \mathbf{J} - \beta\mathbf{A}$ and using $\|\mathbf{g}^{\top} \mathbf{A} \mathbf{b}\|^2 = \text{Tr}(\mathbf{G}_{\boldsymbol{\theta}} \mathbf{A} \mathbf{b} \mathbf{b}^{\top} \mathbf{A}^{\top})$, the conditional MSE
832 writes as a quadratic in β :

$$M(\beta) = \alpha_0 - 2\alpha_1 \beta + \alpha_2 \beta^2, \quad (21)$$

833 with (suppressing the constant prefactor 4)

$$\begin{aligned}\alpha_0 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{J}^\top), \\ \alpha_1 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} \mathbf{A}^\top), \\ \alpha_2 &\triangleq \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top) \mathbf{A}^\top).\end{aligned}$$

834 (We used the symmetry of $\mathbf{G}_\theta$ and $\Sigma_{\mathbf{v}'}$ to consolidate the linear-in- β trace into a single α_1 .) The
835 Hessian is

$$\frac{\partial^2 M}{\partial \beta^2} = 2\alpha_2 = 2 \text{Tr}(\mathbf{G}_\theta \mathbf{A} (\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top) \mathbf{A}^\top) \geq 0,$$

836 since each factor inside the trace is positive semi-definite ($\mathbf{G}_\theta \succeq \mathbf{0}$, $\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top \succeq \mathbf{0}$, and $\mathbf{A}(\Sigma_{\mathbf{v}'} +$
837 $\mathbf{b}\mathbf{b}^\top) \mathbf{A}^\top = \mathbf{M}\mathbf{M}^\top \succeq \mathbf{0}$ for $\mathbf{M} \triangleq \mathbf{A}(\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top)^{1/2}$). Hence $M(\beta)$ is convex in β . The Hessian is
838 strictly positive whenever $\mathbf{G}_\theta^{1/2} \mathbf{M} \neq \mathbf{0}$, which holds outside two pathological regimes: (i) $\mathbf{A} = \mathbf{0}$,
839 equivalently $\mathbf{J} = -\mathbf{I}$ (network with $\partial_{x_t} \mathbf{u}_\theta \equiv \mathbf{0}$, e.g., at saturated initialization), in which case the
840 linear coefficient α_1 also vanishes and $M(\beta)$ is constant; (ii) $\mathbf{g} = \mathbf{0}$ (frozen parameters). Outside
841 these, $M(\beta)$ is *strictly* convex with unique unconstrained minimizer

$$\beta_{\text{matrix}}^* = \frac{\alpha_1}{\alpha_2} = \frac{\text{Tr}(\mathbf{G}_\theta \mathbf{J} \Sigma_{\mathbf{v}'} (\mathbf{J} + \mathbf{I})^\top)}{\text{Tr}(\mathbf{G}_\theta (\mathbf{J} + \mathbf{I})(\Sigma_{\mathbf{v}'} + \mathbf{b}\mathbf{b}^\top)(\mathbf{J} + \mathbf{I})^\top)}. \quad (22)$$

842 Note that β_{matrix}^* may fall outside $[0, 1]$; the box-constrained optimum on the $\beta \in [0, 1]$ interval
843 (cf. equation 7) is then $\text{clip}(\beta_{\text{matrix}}^*, 0, 1)$, attained at the nearer endpoint by convexity.

844 **Step 3: scalar approximation.** Substituting $\mathbf{J} = \kappa \mathbf{I}_d$, $\Sigma_{\mathbf{v}'} = \sigma^2 \mathbf{I}_d$, $\mathbf{A} = (\kappa + 1) \mathbf{I}_d$, and the
845 parameter-isotropy approximation $\mathbf{G}_\theta \propto \mathbf{I}_d$ (which absorbs the parameter-direction normalization
846 into a per-component MSE),

$$\alpha_0 \propto \sigma^2 \kappa^2 d, \quad \alpha_1 \propto \sigma^2 \kappa (\kappa + 1) d, \quad \alpha_2 \propto (\kappa + 1)^2 (\sigma^2 d + \|\mathbf{b}\|^2),$$

847 which equivalently gives

$$M(\beta) \propto \beta^2 (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d ((1 - \beta) \kappa - \beta)^2.$$

848 **Step 4: minimization.** Differentiating and setting to zero (or directly evaluating α_1/α_2),

$$\beta^* = \frac{\alpha_1}{\alpha_2} = \frac{\sigma^2 \kappa (\kappa + 1) d}{(\kappa + 1)^2 (\sigma^2 d + \|\mathbf{b}\|^2)} = \frac{\kappa}{\kappa + 1} \cdot \frac{\sigma^2 d}{\sigma^2 d + \|\mathbf{b}\|^2}.$$

849 **Step 5: optimum value and corner dominance.** At β^* , $(1 - \beta^*) \kappa - \beta^* = \kappa \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$.
850 Substituting back,

$$M(\beta^*) \propto \frac{\sigma^2 d \kappa^2 \|\mathbf{b}\|^2}{\sigma^2 d + \|\mathbf{b}\|^2}.$$

851 For $\kappa > 0$ and $\|\mathbf{b}\|^2 > 0$: $M(\beta^*)/M(0) = \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2) < 1$, so $M(\beta^*) < M(0)$. For the $M(1)$
852 comparison, $M(1) - M(\beta^*) \propto (\kappa + 1)^2 \|\mathbf{b}\|^2 + \sigma^2 d - \sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$, and the first two
853 terms exceed $\sigma^2 d \kappa^2 \|\mathbf{b}\|^2 / (\sigma^2 d + \|\mathbf{b}\|^2)$ since $(\kappa + 1)^2 > \kappa^2$ for $\kappa > 0$. Strict dominance is therefore
854 over the unconstrained interior optimum; if $\beta_{\text{matrix}}^* \notin [0, 1]$ the box-constrained optimum is at the
855 nearer corner and the dominance is non-strict. $\square$

856 B.5 Proof of Proposition 1 (Gradient under the $\beta = 1$ corner loss)

857 **Proposition 1** (Gradient under the $\beta = 1$ corner loss). *The gradient of $\mathcal{L}_{MF}^{EMA}$ takes the form*

$$\nabla_{\theta} \mathcal{L}_{MF}^{EMA} = 2 \mathbb{E}[\mathbf{g} \tilde{\mathbf{r}}^{EMA}],$$

858 where $(\nabla_{\theta} \mathbf{u}_{\theta}) \in \mathbb{R}^{d \times p}$ is the parameter Jacobian, $\tilde{\mathbf{r}}^{EMA} \triangleq \mathbf{V}_{\theta}^{EMA} - \mathbf{v}(\mathbf{x}_t, t) \in \mathbb{R}^d$, and no $\mathbf{J}\mathbf{v}'$ noise
859 term appears.

860 *Proof.* The EMA velocity anchor $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)$ is deterministic given $\mathbf{x}_t$, since the EMA parameters $\bar{\theta}$ are fixed during the gradient step. Under stop-gradient, $V_{\bar{\theta}}^{\text{EMA}}$ depends on θ only through $\mathbf{u}_{\theta}(\mathbf{x}_t, r, t)$. By the same argument as equation 14, the per-step gradient is:

$$\nabla_{\theta} \ell = 2 (\nabla_{\theta} \mathbf{u}_{\theta})^{\top} (V_{\bar{\theta}}^{\text{EMA}} - \mathbf{v}_{\text{cond}}). \quad (23)$$

863 Writing $\mathbf{v}_{\text{cond}} = \mathbf{v}(\mathbf{x}_t, t) + \mathbf{v}'$ and noting that $\tilde{\mathbf{r}}^{\text{EMA}} := V_{\bar{\theta}}^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t)$ is deterministic given $\mathbf{x}_t$:

$$\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} [\nabla_{\theta} \ell] = 2 (\nabla_{\theta} \mathbf{u}_{\theta})^{\top} \tilde{\mathbf{r}}^{\text{EMA}}, \quad (24)$$

864 since $\mathbb{E}[\mathbf{v}' | \mathbf{x}_t] = \mathbf{0}$. The tower property $\nabla_{\theta} \mathcal{L}_{\text{MF}}^{\text{EMA}} = \mathbb{E}_{r, t, \mathbf{x}_t} [\mathbb{E}_{\mathbf{x}_0 | \mathbf{x}_t} [\nabla_{\theta} \ell]]$ gives the stated result. Crucially, no $\mathbf{J}\mathbf{v}'$ term appears because the EMA tangent is deterministic; compare with the original MeanFlow gradient equation 14 where the stochastic tangent produces the $\mathbf{J}\mathbf{v}'$ noise. $\square$

867 B.6 Proof of Proposition 2 (Bias-Variance Tradeoff)

868 **Proposition 2** (Bias-Variance Tradeoff). *The EMA mean-field residual decomposes as*

$$\tilde{\mathbf{r}}^{\text{EMA}} = \mathbf{r}_{\theta} + (t-r) \partial_{\mathbf{x}_t} \mathbf{u}_{\theta} (\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)),$$

869 where $\mathbf{r}_{\theta}$ is the true MeanFlow residual. The FM anchor drives $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$, giving $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$.

871 *Proof.* Expanding $V_{\bar{\theta}}^{\text{EMA}}$ from equation 28, the value of the compound prediction (ignoring the stop-gradient, which does not affect values) is:

$$V_{\bar{\theta}}^{\text{EMA}} = \mathbf{u}_{\theta} + (t-r) (\partial_{\mathbf{x}_t} \mathbf{u}_{\theta} \cdot \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) + \partial_t \mathbf{u}_{\theta}). \quad (25)$$

873 The true MeanFlow residual (with the marginal velocity as tangent) is:

$$\mathbf{r}_{\theta} = \mathbf{u}_{\theta} + (t-r) (\partial_{\mathbf{x}_t} \mathbf{u}_{\theta} \cdot \mathbf{v}(\mathbf{x}_t, t) + \partial_t \mathbf{u}_{\theta}) - \mathbf{v}(\mathbf{x}_t, t). \quad (26)$$

874 Subtracting:

$$\begin{aligned} \tilde{\mathbf{r}}^{\text{EMA}} &= V_{\bar{\theta}}^{\text{EMA}} - \mathbf{v}(\mathbf{x}_t, t) \\ &= \mathbf{r}_{\theta} + (t-r) \partial_{\mathbf{x}_t} \mathbf{u}_{\theta} (\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)). \end{aligned} \quad (27)$$

875 At convergence, the true residual $\mathbf{r}_{\theta} \rightarrow \mathbf{0}$ (the MeanFlow identity is satisfied) and the FM anchor loss equation 29 supervises the boundary condition, driving $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) \rightarrow \mathbf{v}(\mathbf{x}_t, t)$ and hence $\tilde{\mathbf{r}}^{\text{EMA}} \rightarrow \mathbf{0}$. $\square$

878 B.7 Proof of Proposition 3 (Target/Tangent Bias Asymmetry)

879 **Proposition 3** (Target/Tangent Bias Asymmetry). *Let $\hat{\mathbf{v}} = \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)$ be a deterministic proxy with bias $\mathbf{b} \triangleq \hat{\mathbf{v}} - \mathbf{v}(\mathbf{x}_t, t)$. Consider two ways of substituting $\hat{\mathbf{v}}$ into equation 4:*

881 (i) Tangent replacement (the $\beta = 1$ corner): *keep the target $\mathbf{v}_{\text{cond}}$ but replace the JVP tangent. At the loss minimum,*

$$\mathbf{u}^*(\mathbf{x}_t, r, t) - \mathbf{v}(\mathbf{x}_t, t) = -\frac{1}{2}(t-r) \partial_{\mathbf{x}_t} \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2),$$

883 so the stationary error is $\mathcal{O}((t-r) \|\partial_{\mathbf{x}_t} \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at $r \rightarrow t$.

884 (ii) Target replacement: *substitute $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ in the regression target. At $r = t$ the loss minimum satisfies*

$$\mathbf{u}^*(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t) = \mathbf{b},$$

886 i.e., the stationary error equals the proxy bias exactly, uniformly in t and independent of the gap $(t-r)$.

888 *Proof. Tangent replacement.* Substituting $\hat{\mathbf{v}}$ for $\mathbf{v}_{\text{cond}}$ only in the JVP tangent (target unchanged) gives the per-sample compound prediction

$$V_{\bar{\theta}}^{\text{EMA}} = \mathbf{u}_{\theta} + (t-r) (\partial_{\mathbf{x}_t} \mathbf{u}_{\theta} \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}_{\theta}),$$

and the loss is $\mathbb{E}\|V_{\theta}^{\text{EMA}} - v_{\text{cond}}\|^2$. By stationarity, $\mathbb{E}[V_{\theta}^{\text{EMA}} - v_{\text{cond}} \mid \mathbf{x}_t] = \mathbf{0}$ at the minimum. Taking conditional expectation and using $\mathbb{E}[v_{\text{cond}} \mid \mathbf{x}_t] = v(\mathbf{x}_t, t)$,

$$\mathbf{u}^* + (t-r)(\partial_{\mathbf{x}_t} \mathbf{u}^* \cdot \hat{\mathbf{v}} + \partial_t \mathbf{u}^*) = v(\mathbf{x}_t, t).$$

At the same point, the *true* MeanFlow identity (with the marginal velocity as tangent) reads

$$\mathbf{u}^\diamond + (t-r)(\partial_{\mathbf{x}_t} \mathbf{u}^\diamond \cdot \mathbf{v} + \partial_t \mathbf{u}^\diamond) = v(\mathbf{x}_t, t),$$

with $\mathbf{u}^\diamond(\mathbf{x}_t, t, t) = v(\mathbf{x}_t, t)$. Subtracting and writing $\mathbf{u}^* = \mathbf{u}^\diamond + \Delta$,

$$\Delta + (t-r) \partial_{\mathbf{x}_t} \mathbf{u}^* \mathbf{b} + (t-r) D_t \Delta = \mathbf{0}, \quad D_t \Delta \triangleq \partial_t \Delta + v \cdot \partial_{\mathbf{x}_t} \Delta,$$

with the boundary condition $\Delta(\mathbf{x}_t, t, t) = \mathbf{0}$. Substituting $s = t-r$ at fixed r converts this to a first-order linear ODE in s :

$$s \frac{d\Delta}{ds} + \Delta = -s \partial_{\mathbf{x}_t} \mathbf{u}^* \mathbf{b} + \mathcal{O}(\|\mathbf{b}\|^2),$$

which has $s\Delta(s) = -\frac{s^2}{2} \partial_{\mathbf{x}_t} \mathbf{u}^* \mathbf{b} + C$; the boundary condition $\Delta(0) = \mathbf{0}$ forces $C = 0$. Hence

$$\Delta(\mathbf{x}_t, r, t) = -\frac{1}{2}(t-r) \partial_{\mathbf{x}_t} \mathbf{u}^* \mathbf{b} + \mathcal{O}((t-r) \|\mathbf{b}\|^2),$$

i.e., the stationary error is $\mathcal{O}((t-r) \|\partial_{\mathbf{x}_t} \mathbf{u}\| \|\mathbf{b}\|)$ and vanishes at the boundary $r \rightarrow t$. The factor of $\frac{1}{2}$ comes from integrating the material-derivative term, which contributes equally to the leading-order coefficient as the bias source term.

Target replacement. Substituting $\hat{\mathbf{v}}$ for v_{cond} in the regression target yields the loss $\mathbb{E}\|\mathbf{u}_\theta + (t-r) D\mathbf{u}/Dt - \hat{\mathbf{v}}\|^2$. By stationarity, the loss minimum satisfies $\mathbb{E}[\mathbf{u}^* + (t-r) D\mathbf{u}^*/Dt - \hat{\mathbf{v}} \mid \mathbf{x}_t] = \mathbf{0}$. Specializing to $r=t$, the gap term vanishes and the equation collapses to $\mathbf{u}^*(\mathbf{x}_t, t, t) = \hat{\mathbf{v}} = v(\mathbf{x}_t, t) + \mathbf{b}$, so the stationary error at the boundary is exactly $\mathbf{b}$, independent of (t, r) .

Asymmetry. Tangent replacement carries a multiplicative $(t-r)$ factor on the bias and is therefore pinned to zero at the boundary $r=t$ that the FM anchor enforces; target replacement carries no such factor and the bias persists at every t , requiring a boundary regularizer at *every* (r, t) to remove. $\square$

C Implementation Details

To validate the framework empirically (Section 5), we instantiate the $\beta = 1$ corner of Theorem 4 as a backbone-preserving training recipe. Two design choices realize the corner at near-zero compute overhead.

EMA proxy. We take $\hat{\mathbf{v}} = \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)$ where $\bar{\theta}$ is a Polyak-averaged copy of θ [4, 5], $\bar{\theta} \leftarrow \mu \bar{\theta} + (1-\mu)\theta$ with $\mu \in [0, 1)$. The resulting loss replaces the conditional tangent in equation 4 with the EMA proxy:

$$\mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) = \mathbb{E}_{r, t, \mathbf{x}_0, \mathbf{x}_1} \left[\left\| \mathbf{u}_\theta + (t-r) \text{sg}[\text{JVP}(\mathbf{u}_\theta, (\mathbf{x}_t, r, t), (\mathbf{u}_{\bar{\theta}, t}, 0, 1))] - v_{\text{cond}} \right\|_2^2 \right], \quad (28)$$

with $\mathbf{u}_{\bar{\theta}, t} \triangleq \mathbf{u}(\mathbf{x}_t, t, t; \bar{\theta})$. Crucially, the regression target remains v_{cond} : the tangent and target play distinct statistical roles per Section 3.1, and only the tangent needs to be replaced.

Flow-matching anchor. The proxy bias $\mathbf{b} = \mathbf{u}_{\bar{\theta}, t} - v$ must be controlled or it propagates through the $\beta(\mathbf{J} + \mathbf{I})\mathbf{b}$ bias term. We supervise the boundary condition $\mathbf{u}(\mathbf{x}_t, t, t) = v(\mathbf{x}_t, t)$ directly with a small flow-matching loss

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\left\| \mathbf{u}_\theta(\mathbf{x}_t, t-\delta, t) - v_{\text{cond}} \right\|_2^2 \right], \quad (29)$$

where $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$ with $\delta_{\max} \ll 1$. We chose to evaluate the anchor at a small *interior* time offset rather than exactly at the boundary $\delta = 0$ for two reasons: (i) the same $\mathbf{u}(\mathbf{x}_t, t-\delta, t)$ is what the EMA tangent eventually converges to, so the anchor and the JVP target share the same network-evaluation pattern, and (ii) we found training to be slightly more stable when the anchor probes a thin neighborhood of the boundary rather than a single $r = t$ slice. For small δ , $\mathbf{u}(\mathbf{x}_t, t-\delta, t)$ approximates the instantaneous velocity, and equation 1 guarantees $\mathbb{E}[v_{\text{cond}} \mid \mathbf{x}_t] = v(\mathbf{x}_t, t)$, so $\mathcal{L}_{\text{FM}}$ drives $\mathbf{u}_\theta(\mathbf{x}_t, t, t) \rightarrow v(\mathbf{x}_t, t)$, and via EMA, $\mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t) \rightarrow v$ and hence $\mathbf{b} \rightarrow \mathbf{0}$. The full $\beta = 1$ corner loss is

$$\mathcal{L}_{\beta=1}(\theta) = \mathcal{L}_{\text{MF}}^{\text{EMA}}(\theta) + \lambda \mathcal{L}_{\text{FM}}(\theta), \quad (30)$$

Algorithm 1 Training the $\beta = 1$ corner instantiation (EMA tangent + FM anchor)

Require: Dataset $\mathcal{D}$, EMA decay μ , anchor weight λ , anchor interval $[\delta_{\min}, \delta_{\max}]$, learning rate η
Initialize $\bar{\theta} \leftarrow \theta$
for $k = 1, 2, \dots$ **do**
 Sample $(x_0, x_1) \sim \mathcal{D} \times \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$, $t \sim \mathcal{U}[0, 1]$, $r \sim \mathcal{U}[0, t]$, $\delta \sim \mathcal{U}[\delta_{\min}, \delta_{\max}]$
 $x_t \leftarrow (1 - t)x_0 + tx_1$, $v_{\text{cond}} \leftarrow x_1 - x_0$
 $v_{\text{tang}} \leftarrow \text{sg}[\mathbf{u}_{\bar{\theta}}(x_t, t, t)]$ $\triangleright$ EMA tangent (one extra forward pass)
 $V_{\theta} \leftarrow \mathbf{u}_{\theta}(x_t, r, t) + (t - r) \text{sg}[\text{JVP}(\mathbf{u}_{\theta}, (x_t, r, t), (v_{\text{tang}}, 0, 1))]$
 $\ell_{\text{MF}} \leftarrow \|V_{\theta} - v_{\text{cond}}\|_2^2$
 $\ell_{\text{FM}} \leftarrow \|\mathbf{u}_{\theta}(x_t, t - \delta, t) - v_{\text{cond}}\|_2^2$ $\triangleright$ FM anchor (one extra forward pass)
 $\theta \leftarrow \theta - \eta \nabla_{\theta}(\ell_{\text{MF}} + \lambda \ell_{\text{FM}})$
 $\bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu)\theta$ $\triangleright$ EMA update
end for
return θ $\triangleright$ Generate: $\hat{x}_0 = x_1 - \mathbf{u}_{\theta}(x_1, 0, 1)$

927 with $\lambda > 0$. The training procedure is Algorithm 1. Anchor hyperparameters used in our ex-
928 periments: for 2-D toy and DGMM runs, $(\delta_{\min}, \delta_{\max}, \lambda) = (10^{-4}, 10^{-2}, 0.5)$; for the DiT-B/4 /
929 ImageNet-256 $\beta = 1$ run, $(\delta_{\min}, \delta_{\max}, \lambda) = (0, 10^{-3}, 0.1)$.

930 **Optimality.** The combination of EMA proxy plus FM anchor approximates the $\beta \rightarrow 1$ limit
931 of Theorem 4 at the cost of one additional JVP per step. Propositions 1 to 3 (stated and proved
932 in Appendix B) characterize the recipe theoretically: the $\mathbf{J}v'$ amplification of Theorem 2 is elim-
933 inated under the EMA tangent (Proposition 1); the residual EMA bias is controlled by the FM
934 anchor (Proposition 2); and a target/tangent asymmetry justifies why the FM anchor at the boundary
935 $r = t$ alone is sufficient — tangent-replacement bias is multiplicatively damped by $(t - r)$, while
936 target-replacement bias is not (Proposition 3). The same asymmetry locates a failure mode of self-
937 distillation-style variants that use a deterministic proxy for both roles, where the trivial fixed point
938 $\mathbf{u}^* \equiv \hat{v}$ persists without continual external supervision [6, 7].

939 D Additional Results

940 D.1 2-D Toy Benchmark

941 Table 4 reports the matched-method comparison of vanilla MeanFlow ($\beta = 0$) against the full $\beta = 1$
942 corner recipe (EMA tangent + FM anchor; hyperparameters in Appendix C, no trace weight, no
943 adaptive weighting) on six 2-D datasets, averaged over three seeds at 200k optimization steps. The
944 $\beta = 1$ recipe reduces SW_1 on five of six datasets, with the largest reduction (54%) on `swiss_roll`.
945 `two_spirals` is a known low-curvature outlier where vanilla MeanFlow already attains the lowest
946 absolute SW_1 ; the empirical $\beta^* = 0$ on the β -sweep agrees.

947 D.2 Full DGMM β -sweep grid

948 Table 5 reports SW_1 at convergence for every (d, β) pair, with $\text{mean} \pm \text{SEM}$ over three seeds. The
949 bolded cell per row is the empirical $\beta^* = \arg \min_{\beta} \text{SW}_1(\beta)$; the summary view in Table 2 (main
950 text) compares only $\beta = 0$ and β^* per dimension. The grid makes the β -insensitivity of low- d rows
951 visible at full precision: SW_1 is uniform to three decimal places for $d \in \{4, 8, 16\}$.

952 D.3 DiT-B/4 / ImageNet-256 Latent-Space Training

953 We test whether the variance-reduction mechanism of Theorems 2 and 4 scales beyond toys by
954 training DiT [8] variants on ImageNet in the latent space of a frozen Stable Diffusion VAE, follow-
955 ing the recipe of [9]. The four β configurations share identical hyperparameters and code paths;
956 only the loss differs (vanilla MeanFlow MSE for the interior β runs versus the EMA-tangent loss
957 in equation 30 for the $\beta = 1$ corner).

958 **Setup.** Logit-normal (r, t) sampling with mean -0.4 and stddev 1, overlap rate 0.75, AdamW with
959 constant learning rate 10^{-4} , EMA decay 0.9999, batch size 256 across 32 TPU v4 chips. All four
960 configurations ($\beta \in \{0, 0.25, 0.5, 1\}$) share the same DiT-B/4 backbone, data pipeline, optimizer,

Table 4: Sliced Wasserstein distance (lower is better) on the 2-D toy benchmark, averaged over three seeds $\{42, 0, 1\}$ at 200k steps. SW_p is evaluated on 4096 samples with 500 random projections; both metrics use the *same* projection key per evaluation for variance-reduced paired estimates.

Dataset	Metric	MeanFlow ($\beta=0$)	$\beta=1$
checkerboard	SW_1	0.175 ± 0.036	0.134 ± 0.029
	SW_2	0.234 ± 0.042	0.175 ± 0.046
eight_gaussians	SW_1	0.098 ± 0.016	0.085 ± 0.010
	SW_2	0.151 ± 0.022	0.146 ± 0.023
two_moons	SW_1	0.093 ± 0.028	0.072 ± 0.019
	SW_2	0.177 ± 0.068	0.093 ± 0.025
swiss_roll	SW_1	0.098 ± 0.058	0.045 ± 0.003
	SW_2	0.177 ± 0.151	0.060 ± 0.007
two_spirals	SW_1	0.034 ± 0.001	0.037 ± 0.001
	SW_2	0.044 ± 0.001	0.046 ± 0.001
pinwheel	SW_1	0.094 ± 0.011	0.059 ± 0.005
	SW_2	0.128 ± 0.014	0.076 ± 0.005

Table 5: DGMM β -sweep: full grid of mean $\pm$ SEM SW_1 at convergence (200k steps, three seeds $\{42, 0, 1\}$). Bolded cell per row marks the empirical $\beta^* = \arg \min_{\beta} SW_1(\beta)$. The summary view is in Table 2.

d	β										
	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
2	0.152 $\pm$ 0.022	0.151 $\pm$ 0.025	0.149 $\pm$ 0.024	0.148 $\pm$ 0.022	0.149 $\pm$ 0.022	0.150 $\pm$ 0.022	0.150 $\pm$ 0.021	0.148 $\pm$ 0.020	0.148 $\pm$ 0.016	0.148 $\pm$ 0.015	0.146$\pm$0.015
4	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011	0.084 $\pm$ 0.011	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011	0.083$\pm$0.011	0.083 $\pm$ 0.011	0.083 $\pm$ 0.011
8	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062 $\pm$ 0.003	0.062$\pm$0.003
16	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044 $\pm$ 0.002	0.044$\pm$0.003
32	0.034 $\pm$ 0.001	0.034$\pm$0.001	0.037 $\pm$ 0.003	0.038 $\pm$ 0.004	0.041 $\pm$ 0.007	0.041 $\pm$ 0.007	0.035 $\pm$ 0.001	0.034 $\pm$ 0.001	0.036 $\pm$ 0.002	0.037 $\pm$ 0.003	0.036 $\pm$ 0.002
64	0.028 $\pm$ 0.003	0.027 $\pm$ 0.003	0.028 $\pm$ 0.003	0.028 $\pm$ 0.003	0.025$\pm$0.000	0.026 $\pm$ 0.001	0.026 $\pm$ 0.001	0.026 $\pm$ 0.001	0.027 $\pm$ 0.002	0.027 $\pm$ 0.001	0.028 $\pm$ 0.001

EMA schedule, and (r, t) sampling distribution. The interior β runs ($\beta \in \{0, 0.25, 0.5\}$) reuse the baseline configuration verbatim and add only the tangent-mixing rule of equation 7; the $\beta = 1$ corner additionally disables the Karras-style adaptive loss weighting and adds the FM anchor of equation 29 (DiT anchor hyperparameters in Appendix C). The extra EMA forward pass and FM-anchor evaluation per step add up to a $\sim 22\%$ wall-clock overhead on TPU v4 (2.20 steps/s for $\beta = 1$ versus 2.65 steps/s for the baseline); the $\beta \in \{0.25, 0.5\}$ interior runs incur only the EMA forward overhead ($\sim 10\%$).

Four-point matched-step ordering. Table 6 reports FID_{50k} at representative 5k-step checkpoints for all four runs. After the early-training transient ($\leq 25k$ steps), the empirical four-point ordering $FID(\beta = 0) < FID(\beta = 0.25) < FID(\beta = 0.5) < FID(\beta = 1)$ holds at *every* 5k-aligned matched-step checkpoint we logged (54 consecutive datapoints from step 30k to step 295k). The $\beta = 0.25$ vs baseline gap stays positive throughout, ranging from +9.5 early in training (step 30k) to a tight +0.39 at step 295k. The $\beta = 0.5$ vs baseline gap stabilizes around +1.0 $\sim$ +1.5 in the converged regime (steps 200k to 295k), and the $\beta = 1$ vs baseline gap remains +12 $\sim$ +14 across the converged regime.

Converged FID floors and four-point ordering. All four configurations completed their scheduled 300k-step training and have a matched-step eval at step 295k (the baseline crashed at step 239,950 and was resumed from its 240k checkpoint to reach the same horizon as the other three). The four FID floors at step 295k recover the bias-variance ordering of Theorem 4:

$$FID^*(\beta=0) = 11.37 < FID^*(\beta=0.25) = 11.76 < FID^*(\beta=0.5) = 12.51 < FID^*(\beta=1) = 23.36.$$

Quantitatively, taking offsets from the baseline floor 11.37, $\Delta FID(\beta = 0.5) \approx 1.14$ and $\Delta FID(\beta = 1) \approx 11.99$. Anchoring the gradient-MSE prediction $\beta^2 \cdot A$ at the $\beta = 0.5$ point gives $A \approx 4.58$, so the predicted offset at $\beta = 1$ is 4.58 — the empirical 11.99 exceeds this by $\sim 2.6\times$. The FID landscape is therefore *super-linear* in the MSE-axis bias: it penalizes large bias more aggressively than gradient MSE does, which pulls the FID-optimal β past the gradient-MSE interior minimizer to the unbiased

Table 6: DiT-B/4 / ImageNet-256 FID_{50k} (lower is better) at representative 5k-step checkpoints across the four-point β -sweep. All four runs completed their scheduled 300k-step training and have a matched-step eval at step 295k (the baseline crashed at step 239,950 and was resumed from its 240k checkpoint to reach 295k). Bottom rows give the matched-step deltas. The four-point ordering $\text{FID}(\beta=0) < \text{FID}(\beta=0.25) < \text{FID}(\beta=0.5) < \text{FID}(\beta=1)$ holds at every 5k-aligned matched-step checkpoint after the early-training transient (54 consecutive datapoints from step 30k to step 295k).

step (k)	30	50	100	150	200	250	275	295
baseline ($\beta=0$)	128.9	54.5	22.0	15.5	13.0	11.9	11.6	11.37
$\beta=0.25$	138.5	61.6	23.3	16.2	13.3	12.1	11.7	11.76
$\beta=0.5$	145.2	66.9	25.6	17.6	14.4	13.2	12.7	12.51
$\beta=1$	148.8	79.9	41.7	32.1	27.6	25.1	24.1	23.36
Δ ($\beta=0.25$ – baseline)	+9.5	+7.1	+1.3	+0.7	+0.4	+0.2	+0.1	+0.39
Δ ($\beta=0.5$ – baseline)	+16.2	+12.4	+3.6	+2.1	+1.4	+1.3	+1.0	+1.14
Δ ($\beta=1$ – baseline)	+19.9	+25.4	+19.7	+16.6	+14.6	+13.2	+12.5	+11.99

Table 7: Direct DiT-checkpoint measurement of the bias-variance decomposition ingredients of Theorem 4 at $t = 0.5$. The EMA bias $\|\mathbf{b}\|^2$ averages over 128 synthetic-Gaussian $\mathbf{x}_t$ samples and is reported for both methods (which track within 20% of each other). The shrinkage factor uses $\sigma^2 d \approx 8.2 \times 10^3$. The predicted β^* uses $\kappa = 1$ from the Hutchinson trace.

step (k)	$\ \mathbf{b}\ ^2$ (avg)	$\sigma^2 d / (\sigma^2 d + \ \mathbf{b}\ ^2)$	β^*
20	~ 700	0.92	0.46
40	~ 150	0.98	0.49
80	~ 50	0.99	0.50

corner $\beta = 0$. The four-point ordering is the strongest empirical confirmation of the bias-variance decomposition we obtain at DiT scale.

Bias and shrinkage trajectory. To localize the predicted optimum we measure two diagnostics on the DiT checkpoints: an EMA-tracking proxy $\widehat{\|\mathbf{b}\|^2} \triangleq \mathbb{E}_{\mathbf{x}_t} [\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{u}_{\bar{\theta}}(\mathbf{x}_t, t, t)\|^2]$ (the boundary gap between current parameters and the EMA copy), and the irreducible noise floor $\sigma^2 d \approx 8.2 \times 10^3$ from $\text{Tr}(\Sigma_{v'})$ on the same batch. Table 7 reports the trajectory: the EMA-tracking proxy *decays* through training, and the shrinkage factor $\sigma^2 d / (\sigma^2 d + \widehat{\|\mathbf{b}\|^2})$ stays near 1 throughout converged training. We caveat that $\|\mathbf{b}\|^2$ is the EMA-tracking gap, not the true model bias $\|\mathbf{u}_\theta(\mathbf{x}_t, t, t) - \mathbf{v}(\mathbf{x}_t, t)\|^2$; cleanly estimating the latter requires access to the conditional distribution $p(\mathbf{x}_0 | \mathbf{x}_t)$, which is intractable on ImageNet. Combined with $\kappa \approx 1$ from the Frobenius-norm Hutchinson trace, the scalar-isotropic substitution gives $\beta^* \in [0.46, 0.50]$ across the full trajectory; this is a coarse first-cut prediction that the direct matrix-form measurement (Section 5.4) refines to $\beta_{\text{no bias}}^* \approx 0.94$.

Qualitative samples. Figures 5 to 7 show 100 class-conditional generations from the three available checkpoints at step 300k, one figure per β value. The three figures share the same noise key (seed 7) and the same 100 ImageNet class labels (drawn uniformly from $\{0, \dots, 999\}$ with an independent label seed of 7, no curation), so per-position differences across the three figures isolate the effect of the loss alone. The $\beta = 0$ versus $\beta = 1$ contrast is at the perceptual scale, consistent with the $\sim 2\times$ FID gap; the intermediate $\beta = 0.5$ figure is at the metric-difference scale and is not always visually separable from the baseline at this sample budget. The $\beta = 0.25$ checkpoint is absent because that training run finished without persisting any checkpoint (the launch command’s work-dir argument fell back to local-FS semantics due to a URL prefix typo, and FID was computed on-the-fly without an Orbax write); we leave the $\beta = 0.25$ qualitative figure to a future revision in which the run is replayed with corrected logging.

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